Topic 1 Review · IEP · ANSWER KEY

Algebra I · Module 3, Topic 1: Linear Equations and Inequalities · Tuesday, Nov 17 · Test Wed Nov 18

Directions: This review has the same kinds of questions as the Topic 1 test, in the same order. Use the tips and fill in each step. Choose **one** answer for multiple choice. Finish Part 1 (just like the test) first; Part 2 is extra practice. One question per page. You may use the reference box and a calculator. Ask if you need a question read aloud.

Reference box

Solving: distribute → combine like terms → get the variable on one side → undo + and − → undo × and ÷. Check by substituting.

Same sides (3x + 5 = 3x + 5): infinite solutions · **same x-term, different numbers** (3x + 5 = 3x + 2): no solution.

Inequalities: multiply or divide by a negative → **flip** the sign. < or >: open circle · ≤ or ≥: closed circle.

Slope-intercept form: **y = mx + b** · Standard form: **Ax + By = C** · Triangle area = ½bh · Rectangle area = ℓw

Part 1 · Just like the test (Questions 1–11)

1. Jayden has $40 in his bank account. He deposits $15 each week to save for new headphones. Which equation can be used to determine the amount that Jayden has saved, *y*, after *x* weeks?

> **Tip:** Rate × weeks + starting amount.

Step 1: Rate per week = **15**

Step 2: Starting amount = **40**

Step 3: *y* = **15** *x* + **40**

A. $y = 15 + 40x$

B. $y = 15x + 40$

C. $y = -15x + 40$

2. What value of k makes the equation $5 - 4(k + 3) = 9 - 2(2k + 6)$ true?

> **Tip:** Distribute, then combine like terms on each side. Same variable terms + different numbers = no solution.

Step 1: Left side: $5 - 4k - 12 =$ **$-4k - 7$**

Step 2: Right side: $9 - 4k - 12 =$ **$-4k - 3$**

Step 3: $-7 = -3$ is **false**, so: **No solution**

F. 1

G. 2

H. **No solution**

3. What is the solution set for $3(4 - t) > -2t + 15$?

> **Tip:** Dividing or multiplying by a negative **flips** the inequality sign.

Step 1: Distribute: **12 − 3t** $> -2t + 15$

Step 2: Add $2t$, subtract 12: **−t** $>$ **3**

Step 3: Divide by −1 and flip: t **<** **−3**

A. $t > -3$

B. $t > 3$

C. $t < -3$

4. Solve the equation shown.

$0.3(4x - 6) = -1.8x + 4.2$

> **Tip:** Distribute the 0.3 to both terms first.

Step 1: Distribute: **1.2x** − **1.8** $= -1.8x + 4.2$

Step 2: Add $1.8x$ and 1.8: **3** $x =$ **6**

Step 3: $x =$ **2**

Answer: ***x* = 2**

5. Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.

$\frac{1}{3}(12x + 30) = 4(x + 2) + 2$

> **Tip:** Simplify each side. Identical sides = infinite. Same x-term, different numbers = none.

Step 1: Left: $\frac{1}{3}(12x) + \frac{1}{3}(30) =$ **4x + 10**

Step 2: Right: $4x + 8 + 2 =$ **4x + 10**

Step 3: Answer: **infinite solutions**

Answer: **Infinite solutions: both sides simplify to 4x + 10**

6. Maya goes to a school store. She buys 4 notebooks that cost $1.75 each and some pens for $1.20 each. Write an equation for the total cost, y, when Maya buys x pens.

> **Tip:** The thing that changes (pens) gets the x. Fixed items become the constant.

Step 1: Notebooks: 4 × 1.75 = **7.00**

Step 2: Pens: **1.20** x

Step 3: y = **1.20** x + **7.00**

Answer: $y = 1.20x + 7.00$

7. Rewrite the equation in slope-intercept form.

$2x + 5y = 15$

> **Tip:** Get y alone: move the x-term, then divide **every** term by the number in front of y.

Step 1: Subtract $2x$: $5y =$ **$-2x$** $+ 15$

Step 2: Divide every term by **5**

Step 3: $y =$ **$-2/5$** $x +$ **3**

Answer: $y = -\frac{2}{5}x + 3$

8. The formula for the volume of a rectangular prism is $V = \ell wh$. Solve the equation for h.

> **Tip:** Undo what is being done to h. It is multiplied by ℓw, so divide.

Step 1: h is multiplied by ___ℓw___

Step 2: Divide both sides by ___ℓw___

Step 3: h = ___V/(ℓw)___

Answer: $h = \frac{V}{\ell w}$

9. Solve the inequality, and graph the solution on the number line.

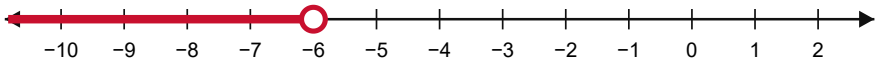
$2.5m - 3 < -1.5(m + 18)$

> **Tip:** Open circle for < or >. Closed circle for ≤ or ≥.

Step 1: Distribute: $2.5m - 3 <$ **$-1.5m - 27$**

Step 2: Add $1.5m$ and 3: **4** $m <$ **-24**

Step 3: m **<** **-6** → **open** circle at −6, shade **left**



Answer: ***m* < −6 (open circle at −6, shade left)**

10. Rewrite the equation $y - 4 = 3(x + 2)$ in slope-intercept form.

> **Tip:** Distribute, then add the number on the y side to both sides.

Step 1: Distribute: $y - 4 =$ **3x + 6**

Step 2: Add **4** to both sides

Step 3: $y =$ **3x + 10**

Answer: **$y = 3x + 10$**

11. A triangle has a height of $(3n - 6)$ inches and a base of 10 inches. A rectangle has a length of 5 inches and a width of $(n + 4)$ inches. The area of the triangle is equal to the area of the rectangle. What is the value of n?

> **Tip:** Triangle area = ½ · base · height. Rectangle area = length · width. Set them equal.

Step 1: Triangle: ½ · 10 · $(3n - 6)$ = **15n − 30**

Step 2: Rectangle: $5(n + 4)$ = **5n + 20**

Step 3: $15n - 30 = 5n + 20 \rightarrow n =$ **5**

Answer: ***n* = 5**

Part 2 · Extra practice (Questions 12–15)

12. Which equation has **no solution**?

> **Tip:** Simplify each one. Look for the same x-term but different numbers.

F. $3(x - 2) = 3x + 5$

G. $3(x - 2) = 3x - 6$

H. $3(x - 2) = 2x + 5$

13. What is the solution set for $-4x + 7 \geq 19$?

> **Tip:** Divide by a negative → flip the sign.

Step 1: Subtract 7: $-4x \geq$ **12**

Step 2: Divide by −4 and flip: x **≤** **−3**

A. $x \geq -3$

B. $x \leq 3$

C. $x \leq -3$

14. The perimeter of a rectangle is $P = 2\ell + 2w$. Solve for w.

> **Tip:** Undo the adding first, then the multiplying.

Step 1: Subtract 2ℓ: 2w = **P − 2ℓ**

Step 2: Divide by **2** : w = **(P − 2ℓ)/2**

Answer: $w = \frac{P - 2\ell}{2}$

15. A gym charges a $25 sign-up fee plus $8 per class. Lena can spend at most $90. Write and solve an inequality for the number of classes, c, she can take.

> **Tip:** “At most” means ≤.

Step 1: $25 + 8c$ ___≤___ 90

Step 2: $8c \leq$ ___65___

Step 3: $c \leq$ ___8.125___ → at most ___8___ classes

Answer: ___**25 + 8c ≤ 90 → c ≤ 8.125, so at most 8 classes**___